

Name:

ID:
Set-A

Section:

```
public class OlympicGames {
    public void startGames() {
        System.out.println("Olympic Games have
started.");
    }

    public void endGames() {
        System.out.println("Olympic Games have ended.");
    }

    private void displayWinner() {
        System.out.println("Winner of the Olympic Games
is not yet decided.");
    }
}

public class SummerOlympics extends OlympicGames
{
    private void displayWinner() {
        System.out.println("Winner of the Summer
Olympics is USA.");
    }

    public void awardMVP() {
        System.out.println("The Most Valuable Player
(MVP) of the Summer Olympics is Michael Phelps.");
    }
}

public class Tokyo2020Olympics extends
SummerOlympics {
    public void awardMVP() {
        System.out.println("The Most Valuable Player
(MVP) of the Tokyo 2020 Olympics is Simone Biles.");
    }
}
```

```
public void hostCity() {

    boolean isTokyo2020OlympicsHosted = true;
    int year = 2020;
    boolean tokyo2020OlympicsHostedInJapan =
false;
    boolean tokyo2020OlympicsHostCityUndecided
= false;

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (i == 2) {
                if (j == 2) {

                    tokyo2020OlympicsHostedInJapan = true;
                }
            }

            else if (i == 1) {
                if (j == 1) {

                    tokyo2020OlympicsHostCityUndecided = true;
                }
            }
        }
    }

    public void awardManOfTheGames() {
        System.out.println("The Man of the Games
for Tokyo 2020 Olympics is Usain Bolt.");
    }
}
```

Consider the following code snippet.

1. **Determine** the Specialization Index (SIX) for the Tokyo2020Olympics. [5]
2. **Illustrate** the Control Flow Graph (CFG) for the hostCity() method, and remember to label the relevant node numbers on your question paper. [7]
3. **Calculate** the cyclomatic complexity of the control flow graph drawn for the hostCity() method. [3]